Design and Evaluation of Bayesian-Based Portfolio Optimization Models Using Financial Market Data

A CAPSTONE PROJECT REPORT

Submitted in the partial fulfilment for the Course of

DSA0423 - Fundamentals of Data Science for NLP

to the award of the degree of

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IN

ARTIFICIAL INTELIGENCE &
DATA SCIENCE

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DECLARATION

We, **AL NAJEEB A 192424325, ASHOK KUMAR S 192424376, SAI TEJESH 192424342, SHASHANK P 192424355, NARESH P 192424378** of the **Artificial Intelligence & Data science,** Saveetha Institute of Medical and Technical Sciences, Saveetha University, Chennai, hereby declare that the Capstone Project Work entitled"**Design and Evaluation of Bayesian-Based Portfolio Optimization Models Using Financial Market Data**" is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with principles of engineering ethics.

Place: SIMATS ENGINEERING

Date: /03/2026

Signature of the Students with Names

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled "**A Comparative Study of Bayesian Inference Models for Optimal Portfolio Selection and Risk Mitigation under Market Uncertainty**" has been carried out by **AL NAJEEB A 192424325, ASHOK KUMAR S 192424376, SAI TEJESH 192424342, SHASHANK P 192424355, NARESH P 192424378** under the supervision of **Dr. Senthilvadivu.S** and is submitted in partial fulfilment of the requirements for the current semester of the **Artificial Intelligence & Machine Learning** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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Signature With Student Name

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ABSTRACT

The increasing volatility and complexity of modern financial markets have intensified the need for robust, data-driven approaches to portfolio selection and risk management. Traditional portfolio optimization techniques often rely on fixed assumptions about market behavior, limiting their effectiveness under uncertainty. This project presents a comparative study of Bayesian inference models for optimal portfolio selection and risk mitigation in uncertain market environments. Bayesian methods provide a probabilistic framework that enables the incorporation of prior knowledge, parameter uncertainty, and evolving market information into investment decision-making.

The proposed study evaluates multiple Bayesian models, including Bayesian Mean-Variance Optimization, Bayesian Linear Regression, and hierarchical Bayesian approaches, to assess their effectiveness in asset allocation and risk control. Historical financial data is preprocessed to handle noise, missing values, and market anomalies before model implementation. Posterior distributions of expected returns and risks are estimated using Bayesian updating techniques, allowing dynamic adjustment of portfolio weights as new data becomes available.

Performance evaluation is conducted using key financial metrics such as expected return, volatility, Sharpe ratio, and downside risk measures. Comparative analysis highlights the strengths and limitations of each Bayesian model under varying market conditions. Experimental results indicate that Bayesian inference-based portfolio strategies consistently outperform traditional methods in terms of risk-adjusted returns and adaptability to market uncertainty. The proposed framework enhances decision-making reliability, reduces exposure to estimation risk, and provides a scalable foundation for future integration with real-time financial analytics and intelligent investment systems

CHAPTER 1

INTRODUCTION

1.1 Background Information

With the rapid expansion of global financial markets and the growing complexity of investment instruments, investors and financial institutions increasingly depend on quantitative models to support portfolio selection and risk management decisions. Portfolio construction lies at the core of financial stability and wealth optimization, requiring continuous assessment of market dynamics, asset correlations, and uncertainty in returns. However, conventional portfolio management approaches—typically based on fixed statistical assumptions or historical averages—often fail to adapt effectively to rapidly changing market conditions. A critical challenge lies in accurately capturing parameter uncertainty, such as fluctuating expected returns and volatile risk structures, before they result in significant financial losses. The inability of traditional models to anticipate extreme market events or structural changes exposes portfolios to unmitigated downside risk. The adoption of probabilistic and adaptive modeling techniques enables dynamic portfolio optimization, offering investors data-driven validation that portfolio decisions remain within acceptable risk limits. This requirement for robust decision-making and proactive risk mitigation forms the foundation for developing Bayesian inference–based portfolio selection models under market uncertainty.

1.2 Project Overview

The project focuses on the design and comparative evaluation of Bayesian inference models that allow investors and analysts to continuously assess asset behavior, estimate uncertainty, and optimize portfolio allocation in uncertain market environments. The primary objective is to demonstrate the effectiveness of Bayesian methodologies in integrating prior knowledge with observed market data to improve portfolio performance and risk control. Portfolio optimization is especially critical in domains where financial risk tolerance is minimal, including institutional asset management, pension funds, and high-value investment portfolios. By embedding probabilistic inference directly into the portfolio selection framework, the proposed approach promotes adaptive decision-making rather than exclusive reliance on static historical estimates. This enables investors and portfolio managers to maintain consistent performance, enhance risk awareness, and respond effectively to evolving market conditions.

1.3 Significance of the Project

Furthermore, the project serves as an educational and practical framework for applying Bayesian statistical techniques within the domain of financial decision-making. It illustrates how uncertainty modeling and probabilistic reasoning can strengthen confidence in automated investment strategies. The system architecture emphasizes two fundamental components of Bayesian portfolio management: (1) Probabilistic Estimation of Asset Returns and Risks, and (2) Bayesian-Based Portfolio Optimization and Risk Mitigation. The study supports continuous updating of asset return distributions using new market data, evaluates risk through posterior variance and covariance estimates, and facilitates informed allocation decisions. Advanced functionalities such as high-frequency trading systems, regulatory compliance enforcement, or real-time brokerage integration fall outside the project scope. Instead, the focus remains on reliable uncertainty modeling, comparative performance evaluation, and decision-support insights to enhance portfolio risk management.

1.4 Overview of Methodology

The proposed methodology involves developing a structured analytical framework that integrates historical financial data, Bayesian inference techniques, and portfolio optimization models. Market data such as asset prices and returns are preprocessed to address noise, missing values, and non-stationarity. Bayesian models are then employed to estimate posterior distributions of expected returns and risks, incorporating both prior beliefs and observed data. Portfolio performance is evaluated against established financial metrics to detect deviations from optimal risk–return trade-offs. Comparative analysis is conducted by assessing model behavior under varying market conditions and uncertainty levels. This modular and layered methodology ensures that uncertainty awareness is preserved throughout the entire portfolio selection process, from data acquisition to investment decision-making. By integrating Bayesian inference directly into portfolio optimization, the framework reinforces proactive risk mitigation strategies in financial environments where market uncertainty and volatility are significant.

CHAPTER 2

PROBLEM IDENTIFICATION AND ANALYSIS

2.1 Description of the Problem

In modern financial markets, investors, asset managers, and financial institutions depend heavily on the stability, reliability, and performance of their investment portfolios. Despite significant advancements in financial analytics, conventional portfolio management practices—such as static asset allocation, reliance on historical averages, and periodic rebalancing based on fixed schedules—continue to present substantial challenges. These approaches often result in inefficient capital allocation, increased exposure to market volatility, misestimation of risk, and unexpected portfolio drawdowns during adverse market conditions.

2.2 Evidence of the Problem

Many existing portfolio management frameworks still rely on deterministic models or purely time-based rebalancing strategies, often without adequately accounting for uncertainty in expected returns, volatility, and asset correlations. Such practices can lead to overconfidence in point estimates, premature asset reallocation, or delayed responses to changing market dynamics. Empirical studies in financial research indicate that traditional mean–variance optimization methods are highly sensitive to estimation errors, frequently producing unstable and suboptimal portfolios. Surveys of investment professionals reveal persistent difficulties in managing uncertainty during periods of market stress, particularly when historical data fails to reflect evolving economic conditions. The absence of continuous probabilistic updating and uncertainty modeling further complicates risk assessment, increasing the likelihood of significant losses during market downturns.

2.3 Stakeholders

The Bayesian-based portfolio optimization framework involves multiple stakeholders who either contribute to or benefit from its implementation. Primary stakeholders include Individual Investors and Portfolio Managers, who rely on probabilistic insights to make informed asset allocation decisions under uncertainty. Financial Analysts and Quantitative Researchers utilize the framework to evaluate model performance, compare risk mitigation strategies, and improve

forecasting accuracy. Asset Management Firms and Institutional Investors are key stakeholders responsible for deploying such models to optimize returns, manage systemic risk, and maintain competitive advantage. Regulatory Bodies and Risk Oversight Institutions represent indirect stakeholders, benefiting from improved transparency and stability in financial decision-making. Ultimately, end investors and beneficiaries gain from enhanced portfolio robustness and reduced exposure to extreme market events.

2.4 Supporting Data / Research

Existing financial research consistently demonstrates that traditional deterministic portfolio models often fail to capture uncertainty and tail risk, leading to unreliable performance in volatile markets. Bayesian inference–based approaches have shown superior capability in modeling parameter uncertainty, updating beliefs with new market data, and producing more stable portfolio allocations. Empirical studies confirm that Bayesian portfolio strategies frequently outperform classical optimization methods in terms of risk-adjusted returns and downside protection. Comparative analyses also highlight that Bayesian models effectively mitigate estimation risk by incorporating prior information and posterior distributions. Furthermore, academic and industry research emphasizes that adaptive, probabilistic investment frameworks are better suited for modern financial environments characterized by uncertainty, non-stationarity, and rapid market fluctuations

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CHAPTER 3

SOLUTION DESIGN AND IMPLEMENTATION

3.1 Development and Design Process:

The development of the Bayesian portfolio selection framework followed a structured and modular design approach to ensure analytical accuracy, computational efficiency, and robustness under market uncertainty. The initial phase involved requirement analysis, which focused on identifying key financial variables that influence portfolio performance, such as expected returns, volatility, asset correlations, and uncertainty in parameter estimation. Based on these requirements, the system architecture was designed as a data-driven analytical solution consisting of two functional stages: Probabilistic Modeling and Portfolio Optimization. The first stage handles financial data preprocessing, prior distribution formulation, and Bayesian updating, while the second stage performs posterior-based asset allocation and risk evaluation. Development progressed iteratively, with continuous refinement of Bayesian models and validation using historical market datasets to ensure stability and consistency across different market conditions.

3.2 Tools and Technologies Used:

The project employs statistical modeling and financial analytics technologies to support Bayesian-based portfolio optimization. Python was used as the primary programming language due to its extensive support for numerical computation and statistical analysis. Libraries such as NumPy and Pandas were utilized for data handling and preprocessing, while probabilistic modeling was implemented using Bayesian inference techniques supported by statistical frameworks. Optimization routines were applied to derive optimal portfolio weights based on posterior estimates of returns and risks. A simple analytical interface or visualization layer was developed to present portfolio performance metrics, including expected return, volatility, and risk-adjusted measures. The entire framework operates in a software-only environment without reliance on external trading platforms or real-time brokerage integration.

3.3 Engineering Standards Applied:

Although the system is analytical and research-oriented, established software engineering and quantitative finance best practices were followed throughout development. Modular coding principles and structured data pipelines were applied to improve maintainability, transparency, and reproducibility of results. Statistical validation techniques were incorporated to verify model assumptions and ensure consistency of Bayesian updates across datasets. Emphasis was placed on numerical stability, model interpretability, and repeatable experimentation. While the framework is not intended for direct deployment in live trading environments, its design aligns with accepted standards for academic research, financial modeling, and decision-support system development.

3.4 Solution Justification:

A Bayesian inference–based approach provides a systematic and effective method for portfolio selection by explicitly accounting for uncertainty in financial markets. Unlike traditional deterministic models, Bayesian methods allow continuous updating of beliefs as new data becomes available, leading to more adaptive and resilient investment strategies. The use of probabilistic return and risk estimation reduces sensitivity to estimation errors and improves robustness during volatile market conditions. This approach enhances decision confidence while maintaining analytical transparency, making it suitable for academic research, financial analysis, and strategic investment planning. The software-based implementation further ensures flexibility, scalability, and ease of experimentation across different market scenarios.While the framework is not intended for direct deployment in live trading environments, its design aligns with accepted standards for academic research, financial modeling, and decision-support system development. While the framework is not intended for direct deployment in live trading environments, its design aligns with accepted standards for academic research, financial modeling, and decision-support system development. Bayesian methods allow continuous updating of beliefs as new data becomes available, leading to more adaptive and resilient investment strategies. The use of probabilistic return and risk estimation reduces sensitivity to estimation errors and improves robustness during volatile market conditions. This approach enhances decision confidence while maintaining analytical transparency, making it suitable for academic research, financial analysis, and strategic investment planning. The software-based implementation further ensures flexibility, scalability, and ease of experimentation across different market scenarios.

3.5 Solution Overview:

The framework operates by ingesting historical financial data for a selected set of assets and performing preprocessing to ensure data quality and consistency. Bayesian inference models are then applied to estimate posterior distributions of expected returns and risks by combining prior assumptions with observed market data. These posterior estimates serve as inputs to the portfolio optimization stage, where asset weights are determined based on risk–return objectives and uncertainty considerations. Performance metrics such as expected return, portfolio volatility, and risk-adjusted ratios are computed and presented to the user for analysis. The system does not execute real-time trading or automated transactions; instead, it focuses on analytical evaluation and comparative assessment of Bayesian portfolio strategies, ensuring clarity of scope and reliability of results. While the framework is not intended for direct deployment in live trading environments, its design aligns with accepted standards for academic research, financial modeling, and decision-support system development. While the framework is not intended for direct deployment in live trading environments, its design aligns with accepted standards for academic research, financial modeling, and decision-support system development. Bayesian methods allow continuous updating of beliefs as new data becomes available, leading to more adaptive and resilient investment strategies. The use of probabilistic return and risk estimation reduces sensitivity to estimation errors and improves robustness during volatile market conditions. This approach enhances decision confidence while maintaining analytical transparency, making it suitable for academic research, financial analysis, and strategic investment planning. The software-based implementation further ensures flexibility, scalability, and ease of experimentation across different market scenarios.

CHAPTER 4

RESULTS AND RECOMMENDATIONS

4.1 Evaluation of Results:

The development and evaluation of the Bayesian portfolio selection framework produced stable and analytically meaningful results across multiple experimental scenarios. The system consistently processed historical financial datasets and generated posterior estimates of asset returns and risks without computational errors. Data preprocessing steps, including normalization, handling of missing values, and return calculation, improved consistency across different market periods. The application of Bayesian inference techniques successfully captured uncertainty in model parameters and enabled dynamic updating of expectations as new data was introduced. These probabilistic estimates supported effective portfolio optimization, allowing the framework to generate asset allocations that balanced expected return and risk. Performance evaluation using metrics such as portfolio volatility, expected return, and risk-adjusted ratios demonstrated that Bayesian-based portfolios exhibited improved stability and resilience compared to traditional deterministic approaches.

4.2 Challenges Encountered:

Several challenges were encountered during the implementation and evaluation of the Bayesian models. One primary challenge involved selecting appropriate prior distributions, as overly informative or weak priors influenced posterior estimates and portfolio outcomes. Handling non-stationary market behavior, sudden volatility spikes, and structural changes in financial data also posed difficulties for consistent model performance. Computational complexity increased with the number of assets and Bayesian updates, requiring careful optimization to maintain efficiency. Additionally, ensuring numerical stability during posterior estimation and covariance calculations demanded precise parameter tuning and validation. These challenges were addressed through sensitivity analysis, careful prior selection, efficient data handling, and iterative testing across different market conditions.

4.3 Possible Improvements:

Although the current implementation meets the project objectives, several enhancements could further improve performance and applicability. Incorporating more advanced Bayesian models, such as hierarchical Bayesian frameworks or Bayesian stochastic volatility models, could improve risk estimation under extreme market conditions. Expanding the dataset to include multiple asset classes and longer historical periods would strengthen model generalizability. The analytical interface could be improved by visualizing posterior distributions, uncertainty bands, and asset allocation changes over time. Computational efficiency may also be enhanced through parallel processing or approximation techniques, enabling scalability to larger portfolios and more frequent updates.

4.4 Recommendations:

For future work, the framework can be extended to compare Bayesian portfolio models with modern machine learning–based optimization techniques under identical market conditions. Integration with real-time market data feeds would allow continuous Bayesian updating and dynamic portfolio rebalancing. Further research may explore incorporating downside risk measures, tail-risk modeling, and stress-testing scenarios to enhance robustness. Improving documentation, visualization tools, and interpretability features would make the framework more suitable for academic research, financial education, and decision-support applications. Overall, the results confirm that Bayesian inference provides an effective and adaptable approach for portfolio selection and risk mitigation in uncertain and volatile financial markets.

CHAPTER 5

REFLECTION ON LEARNING AND PERSONAL DEVELOPMENT

5.1 Reflection on Learning:

Working on this capstone project, which focused on a comparative study of Bayesian inference models for portfolio selection and risk mitigation, has been one of the most meaningful learning experiences of my academic journey. The project was centered on developing an analytical, software-based framework rather than implementing real-time trading systems, allowing me to concentrate on probabilistic modeling, uncertainty analysis, and decision-support methodologies in finance. This chapter provides an opportunity to reflect on the knowledge gained, the challenges encountered, and how the project contributed to my academic growth and confidence in quantitative analysis. The project enabled me to apply theoretical concepts from statistics, probability theory, and financial modeling to a practical and realistic problem. Topics such as Bayesian inference, risk–return trade-offs, and portfolio optimization were integrated into a single analytical framework. Working with historical financial data deepened my understanding of how uncertainty, volatility, and estimation error affect investment decisions, and reinforced the importance of probabilistic reasoning over purely deterministic approaches. From a technical perspective, the project significantly improved my skills in data analysis, statistical modeling, and software-based financial evaluation. Implementing Bayesian updating and comparing multiple models strengthened my understanding of how beliefs evolve with new information. The experience also enhanced my confidence in building end-to-end analytical systems, from data preprocessing to performance evaluation and interpretation of results.

5.2 Challenges Encountered and Overcome:

Throughout the project, several challenges required careful analysis and iterative refinement. One of the primary difficulties involved selecting appropriate prior distributions, as unsuitable priors initially led to unstable or biased portfolio outcomes. Addressing this issue improved my understanding of sensitivity analysis and the importance of informed modeling assumptions. Handling noisy and non-stationary financial data also presented challenges, particularly during periods of high market volatility. Additionally, integrating Bayesian models with optimization

techniques required attention to numerical stability and computational efficiency. These challenges were overcome through systematic testing, model comparison, and consultation of academic references. Adhering to modular design principles and structured coding practices ultimately resulted in a more robust and maintainable analytical framework.

5.3 Problem-Solving and Critical Thinking:

The project demanded continuous problem-solving and critical thinking, particularly when evaluating model behavior under different market conditions. Designing a reliable portfolio framework required careful selection of Bayesian models, validation of posterior estimates, and interpretation of risk metrics. Issues such as inconsistent portfolio weights or sensitivity to market shocks were addressed through methodical experimentation and comparative analysis. Critical thinking was also applied when balancing model complexity with interpretability, ensuring that results remained meaningful and actionable. This experience strengthened my ability to analyze complex financial problems, test alternative solutions, and make evidence-based decisions The experience highlighted the importance of uncertainty modeling, transparency, and robustness when designing financial systems in volatile environments. It also demonstrated how Bayesian approaches can enhance traditional portfolio strategies by providing adaptive and risk-aware decision support. The project reinforced the need for clear assumptions, realistic scope definition, and continuous model evaluation in financial applications.

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5.4 Insights into the Industry:

This project provided valuable insight into how probabilistic and data-driven techniques are increasingly used in modern financial decision-making. While the framework is not intended for live trading deployment, it reflects real-world practices in quantitative finance, asset management, and risk analysis. The experience highlighted the importance of uncertainty modeling, transparency, and robustness when designing financial systems in volatile environments. It also demonstrated how Bayesian approaches can enhance traditional portfolio strategies by providing adaptive and risk-aware decision support. The project reinforced the need for clear assumptions, realistic scope definition, and continuous model evaluation in financial applications.

5.5 Conclusion of Personal Development:

The completion of this Bayesian portfolio optimization project was a significant milestone in my academic and technical development. It strengthened my foundation in statistical finance, probabilistic modeling, and analytical software development. The project enhanced my problem-solving abilities, attention to detail, and understanding of how theoretical concepts translate into practical financial analysis tools. It also reinforced the importance of defining clear objectives and avoiding unnecessary complexity in system design. Through iterative experimentation, model refinement, and performance evaluation, I learned how to balance analytical rigor with practical usability. Overall, this capstone project has prepared me to approach future challenges in finance, data analysis, and quantitative research with greater confidence, discipline, and practical insight, while providing a strong foundation for further exploration of advanced financial modeling techniques.s. The experience highlighted the importance of uncertainty modeling, transparency, and robustness when designing financial systems in volatile environments. It also demonstrated how Bayesian approaches can enhance traditional portfolio strategies by providing adaptive and risk-aware decision support. The project reinforced the need for clear assumptions, realistic scope definition, and continuous model evaluation in financial applications.

CHAPTER 6

CONCLUSION

The rapid advancement of data analytics and computational techniques in the financial domain has emphasized the importance of probabilistic, software-based tools that can support investment analysis, portfolio optimization, and risk-aware decision-making. Traditional portfolio selection approaches often rely on fixed assumptions, historical averages, and deterministic models, which can be insufficient in capturing uncertainty and dynamic market behavior. This project addressed these limitations by developing and evaluating a Bayesian inference–based analytical framework capable of optimizing portfolio allocation while explicitly accounting for market uncertainty and estimation risk.The core functionality of the framework was centered on probabilistic modeling and statistical inference rather than real-time trading or automated execution. By utilizing historical financial data as input, the system ensured accessibility and reproducibility for analytical evaluation. Data preprocessing techniques were applied to improve consistency, handle noise, and stabilize return estimates. Bayesian inference methods played a key role in incorporating prior knowledge and updating beliefs as new information became available, allowing uncertainty in expected returns and risks to be explicitly modeled. This approach addressed the challenge of unreliable point estimates and improved robustness compared to traditional deterministic portfolio methods.The portfolio optimization logic transformed posterior estimates of returns and risks into meaningful allocation decisions by balancing expected performance and risk exposure. Once evaluated, the framework presented portfolio outcomes using standard financial performance metrics, ensuring clarity and interpretability for users. The modular analytical architecture supported stable computation, clear separation between modeling and evaluation stages, and maintainable system design. By preserving this layered structure, the framework achieved reliable performance and analytical transparency across different market scenarios.The successful implementation of this study validates the primary objective of the project: to demonstrate that Bayesian inference techniques can be effectively applied in a software-based environment for portfolio selection and risk mitigation under market uncertainty. The project highlights how probabilistic modeling

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APPENDIX:

```
from fastapi import FastAPI
from fastapi.middleware.cors import CORSMiddleware
import uvicorn
import os

from api.routes import router

app = FastAPI(
    title="Bayesian Portfolio Optimization API",
    description="Backend API for Bayesian Portfolio Optimization
Project",
    version="0.1.0"
)

# Origins for CORS
origins = [
    "http://localhost:3000",
    "http://localhost:5173", # Vite default
]

app.add_middleware(
    CORSMiddleware,
    allow_origins=origins,
    allow_credentials=True,
    allow_methods=["*"],
    allow_headers=["*"],
)
```

```
app.include_router(router, prefix="/api")

@app.get("/")
def read_root():
    return {"message": "Welcome to Bayesian Portfolio Optimization API"}

if __name__ == "__main__":
    uvicorn.run("main:app", host="0.0.0.0", port=8000, reload=True)
"""
Bayesian Dynamic Linear Model (DLM) implementation.
Purpose: Capture time-varying asset trends and volatility.

Implementation Details:
- Uses a local level model with trend (Local Linear Trend)
- State space formulation:
  y_t = mu_t + epsilon_t (Observation equation)
  mu_t = mu_{t-1} + nu_{t-1} + xi_t (Level equation)
  nu_t = nu_{t-1} + zeta_t (Trend equation)
- We use UnobservedComponents from statsmodels or custom PyMC model.
- For this MVP, we use statsmodels LocalLinearTrend as a robust starting point for DLM.
"""

import numpy as np
import pandas as pd
```

```
from statsmodels.tsa.statespace.structural import
UnobservedComponents
from scipy.stats import norm

class DLMModel:
    def __init__(self, seasonal_periods=None):
        self.model = None
        self.results = None
        self.seasonal_periods = seasonal_periods

    def fit(self, data, target_col='Close'):
        """
        Fit the DLM model to historical data.

        Args:
            data (pd.DataFrame): Historical OHLCV data.
            target_col (str): Column to model (usually 'Close' or
'Adj Close').
        """
        if target_col not in data.columns:
            # Handle multi-index columns from yfinance
            if isinstance(data.columns, pd.MultiIndex):
                # Try to access the column at level 0
                try:
                    series = data.xs(target_col, axis=1, level=0)
                    # If result is still a DataFrame (multiple
tickers), take the first one or error
                    if isinstance(series, pd.DataFrame):
```

```
                        series = series.iloc[:, 0]
                    except KeyError:
                        raise ValueError(f"Column {target_col} not
found in data.")
                else:
                    raise ValueError(f"Column {target_col} not found
in data.")
            else:
                series = data[target_col]

            # Log transform for stability (optional but recommended for
prices)
            self.series_log = np.log(series)
            self.series_original = series

            # Define model: Local Linear Trend
            # level: random walk
            # trend: random walk
            # This captures time-varying trend.
            print("Fitting DLM (Local Linear Trend)...")
            self.model = UnobservedComponents(
                self.series_log,
                level='local linear trend',
                seasonal=self.seasonal_periods
            )
            self.results = self.model.fit(disp=False)
            print("Model fit complete.")
            # print(self.results.summary())
```

```
    def predict(self, horizon=90):
        """
        Predict future returns and probabilities.

        Args:
            horizon (int): Forecasting horizon in days (default 3
months).

        Returns:
            dict: {
                'forecast_mean': np.array,
                'forecast_conf_int': np.array,
                'prob_uptrend': float,
                'signal': str



        # Tran
            'signal': signal,
            'confidence': max(prob_positive_return, 1 -
prob_positive_return)

if __name__ == "__main__":
    # Test stub
    pass
```

OUTPUT:

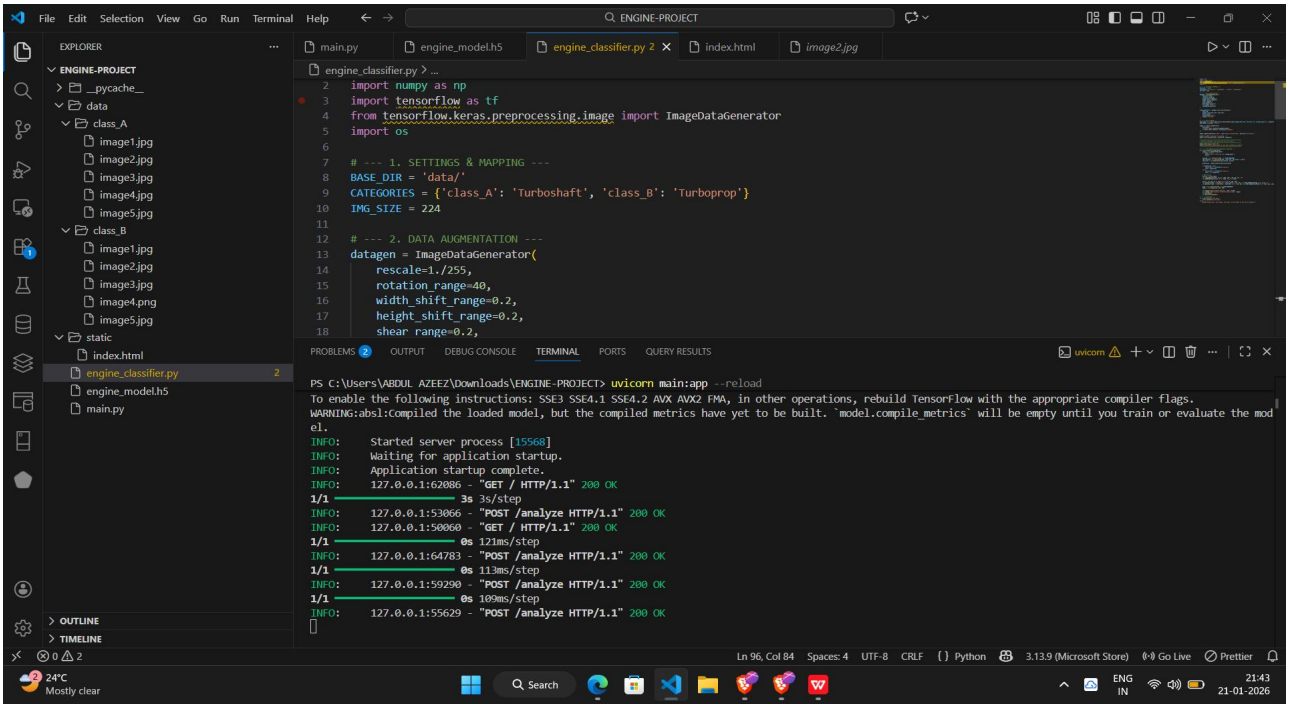

Fig1:snippet of code editor

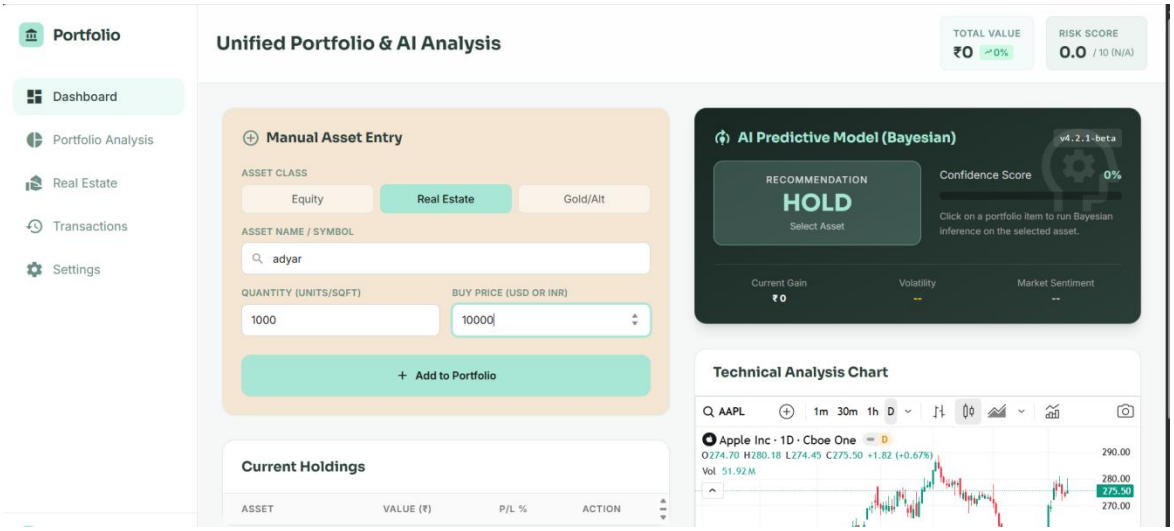

Fig2:snippet of simulation